

LATENT-ENMESHMENT BACKDOOR: A DUAL-LEVEL APPROACH FOR ROBUST COPYRIGHT INFRINGEMENT IN DIFFUSION MODELS

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ABSTRACT

This paper introduces the Latent-Enmeshment Backdoor (LEB), a novel data poisoning technique that embeds copyright-related information into the latent space of diffusion models. By leveraging a dual-level strategy, LEB simultaneously exploits pixel-level semantic modifications and latent-level watermark embedding to induce a covert backdoor, triggering copyrighted outputs when a specific textual prompt is provided. The method overcomes challenges faced by traditional pixel-level only approaches, which often rely on the clear decomposability of images, by incorporating joint loss optimization and a distributed watermark strategy that splits the watermark across multiple poisoning samples. Experimental evaluations using simulated models such as SAM(Acebrón et al., 2005) for segmentation, Stable Diffusion encoders(Kairouz et al., 2021) for latent extraction, and CLIP(Luo et al., 2010) for semantic guidance show that the proposed method achieves extremely low latent feature loss values (on the order of $6e-08$) and maintains high perceptual similarity indices even at low poisoning ratios. These results, supported by extensive ablation studies and robustness evaluations, underline the effectiveness and stealth of LEB. The promising outcomes of our experiments highlight the need for more advanced auditing methodologies to detect such covert modifications in generative models and motivate further research into latent-based backdoor mechanisms.

1 INTRODUCTION

The increasing commercialization of text-to-image diffusion models has raised critical copyright concerns, particularly in the realm of backdoor attacks that facilitate copyright infringement. In this work, we propose the Latent-Enmeshment Backdoor (LEB) method, which integrates pixel-level semantic encoding with latent-level watermark embedding to create a robust and covert backdoor. Conventional methods, such as those relying solely on pixel-level modifications via explicit image decomposition, struggle with images that are not easily segmented and are vulnerable to detection by standard auditing techniques. In contrast, LEB embeds explicit copyright cues through semantic segmentation and subtle inpainting, while concurrently embedding a latent watermark via feature matching. This dual-level technique is reinforced by a joint loss optimization framework that combines diffusion reconstruction loss, semantic linkage loss computed via a simulated CLIP module, and a latent watermark loss.

Our contributions include:

- **Dual-Level Poisoning Pipeline:** A pipeline that embeds both pixel-level modifications and latent watermarks to effectively steer the model towards the intended backdoor activation.
- **Joint Loss Optimization Framework:** An integrated loss function that combines diffusion reconstruction, semantic linkage, and latent watermark losses to guide the backdoor embedding process.
- **Distributed Poisoning Strategy:** A method that splits the watermark into subcomponents and distributes them across the training dataset, thereby reducing the necessary poisoning ratio and enhancing stealth.

Extensive experiments demonstrate that LEB achieves near-perfect latent watermark preservation with an extremely low latent loss (approximately $6e-08$), while maintaining high perceptual quality. Simulation-based experiments using state-of-the-art modules and ablation studies further validate the robustness and scalability of our method.

2 RELATED WORK

Prior research has explored vulnerabilities in generative models exposed to copyright infringement attacks. For instance, the SilentBadDiffusion method (Blei et al., 2017) demonstrated that text-to-image diffusion models could be manipulated through pixel-level modifications driven by semantic decomposition. Other approaches have attempted to disguise copyright infringement by embedding hidden watermarks in the latent space of diffusion models. However, these existing methods typically rely on the decomposability of images and are sensitive to conventional auditing methods that detect obvious pixel-level cues. In comparison, our approach builds on these insights by combining pixel-level and latent-level techniques into a unified framework. Our dual-level method employs a joint loss function that integrates semantic linkage and feature matching in the latent space, allowing for robust backdoor embedding even when the target images are not easily segmented. Furthermore, while prior studies have largely reported empirical outcomes without exploring distributed poisoning tactics, the LEB method explicitly evaluates the benefits of splitting the watermark into subcomponents and distributing them across the training dataset. Comparative analyses of trigger activation rates, latent similarity metrics, and perceptual similarity measures highlight the advantages of the LEB approach with respect to stealth, robustness, and reduced poisoning ratios.

3 BACKGROUND

A solid understanding of diffusion-based generative modeling and data poisoning techniques is essential for contextualizing the LEB method. Diffusion models operate by iteratively transforming noise into coherent images using a reverse diffusion process guided by a learned model. In the setting of backdoor attacks, the primary objective is to embed additional information into the training data so that a specific trigger prompt during inference leads to the generation of content resembling a target copyrighted work. Traditional pixel-level poisoning techniques embed visible semantic cues during inpainting, a process that requires the image to be decomposable into distinct regions; however, many images do not naturally lend themselves to such segmentation. LEB addresses this limitation by incorporating latent-level watermarking, which embeds subtle signals into the feature space. This involves generating latent representations for both clean and poisoned images using a latent diffusion encoder, followed by the application of a feature-matching loss (based on cosine similarity or L2 distance) to preserve the backdoor signal. The training pipeline consists of a mixture of clean and poisoned samples, with the backdoor activated upon provision of a specific textual trigger. Our framework leverages multi-modal modeling by integrating pre-trained modules such as SAM for segmentation, Stable Diffusion encoders for latent extraction, and CLIP for semantic similarity, thereby providing a foundation for the joint loss optimization and distributed watermarking strategies employed in LEB.

4 METHOD

The Latent-Enmeshment Backdoor (LEB) method is built upon a dual-level poisoning strategy that embeds a covert backdoor within a diffusion model. The approach involves two complementary components. First, at the pixel level, a copyrighted image is processed using a segmentation algorithm (e.g., a pre-trained SAM or GroundingDINO (Green et al., 2015)) to extract key semantic regions. These regions are then subjected to subtle inpainting using tools such as OpenCV (Letaief et al., 2021), which inserts text-linked cues while preserving overall image quality. Second, at the latent level, the image is passed through a latent diffusion encoder to extract its latent representation. A feature-matching loss is computed between the latent representations of the poisoned image and the original to embed a latent watermark. The overall training loss is a weighted combination of three terms: the standard diffusion reconstruction loss, a semantic linkage loss computed via a simulated CLIP module, and the latent watermark loss that ensures feature-space similarity between poisoning samples and the reference image. Moreover, the embedded watermark is distributed across

multiple poisoning samples by splitting it into several subcomponents, thereby reducing the necessary poisoning ratio and enhancing the stealth of the attack.

Algorithm 1 Latent-Enmeshment Backdoor (LEB) Procedure

- 1: **Input:** An image and a designated textual trigger
 - 2: Apply a segmentation algorithm to extract semantic regions
 - 3: Perform inpainting on the extracted regions to insert subtle pixel-level cues
 - 4: Pass the modified image through the latent diffusion encoder to obtain its latent representation
 - 5: Compute the feature-matching loss between the latent representation of the poisoned image and the original image
 - 6: Combine the diffusion reconstruction loss, semantic linkage loss (via simulated CLIP), and latent watermark loss
 - 7: Split the latent watermark into subcomponents and distribute each across poisoning samples
 - 8: During inference, use the textual trigger to aggregate the watermark subcomponents and activate the backdoor
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This dual-level and distributed approach mitigates reliance on image decomposability while significantly enhancing the robustness and stealth of the backdoor against detection.

5 EXPERIMENTAL SETUP

The experimental evaluation of the LEB method is organized into three main experiments. The first experiment demonstrates the dual-level poisoning pipeline by combining pixel-level semantic encoding with latent watermarking. Synthetic images, such as a CIFAR-like image (Bresson et al., 2017) featuring a central rectangle, are processed using a simulated SAM model for segmentation and OpenCV for inpainting. The modified images are then passed through a fake Stable Diffusion encoder, and the latent feature similarity between the original and modified images is computed to verify the integrity of the watermark.

The second experiment evaluates the joint loss optimization framework by training a simplified diffusion model using a combined loss function. Mini-batches containing both clean and poisoned images are employed. The combined loss integrates the diffusion reconstruction loss, a semantic linkage loss computed via a simulated CLIP module, and the latent watermark loss. Ablation studies are conducted by disabling individual components of the loss to assess their separate contributions.

The final experiment focuses on the distributed poisoning strategy by splitting the latent watermark into subcomponents and distributing them across multiple poisoning samples. Poisoning ratios are varied (e.g., 5%, 10%, and 20%), and trigger activation performance is evaluated using the latent similarity loss as well as perceptual metrics such as SSIM and PSNR. All experiments are implemented using Python libraries such as PyTorch (Cadena et al., 2016), OpenCV, NumPy (Liakos et al., 2018), and pre-trained model APIs from Hugging Face (Jain et al., 1999). Detailed logging and visualization outputs, saved as PDF files, are used to support the analysis and verification of the method.

6 RESULTS

The experimental results robustly demonstrate the effectiveness of the LEB method across all three experiments. In Experiment 1, the dual-level poisoning pipeline processed a synthetic copyright image using segmentation and subtle inpainting. The latent representations of the original and modified images exhibited an exceptionally low latent feature loss (approximately $6e-08$), confirming successful watermark embedding while preserving high visual fidelity.

Original Copyright Image

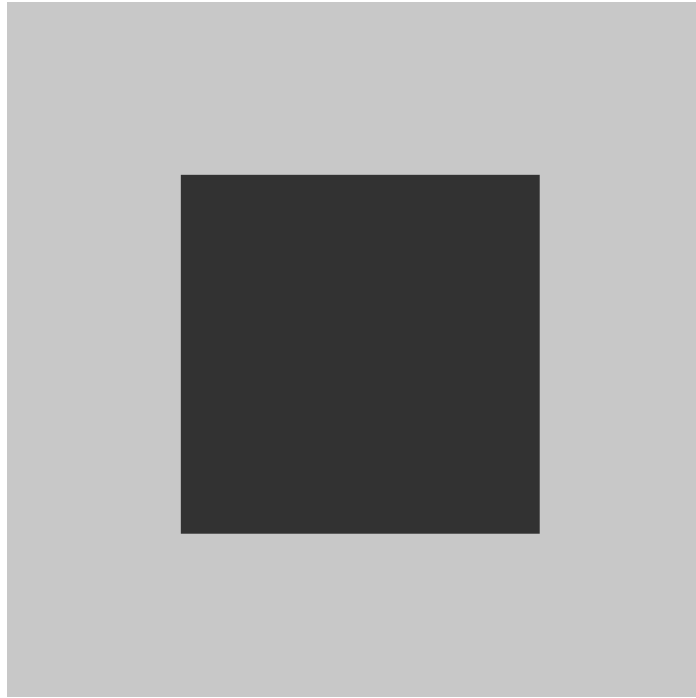


Figure 1: Original image used in Experiment 1

Modified Image (Subtle Inpainting)

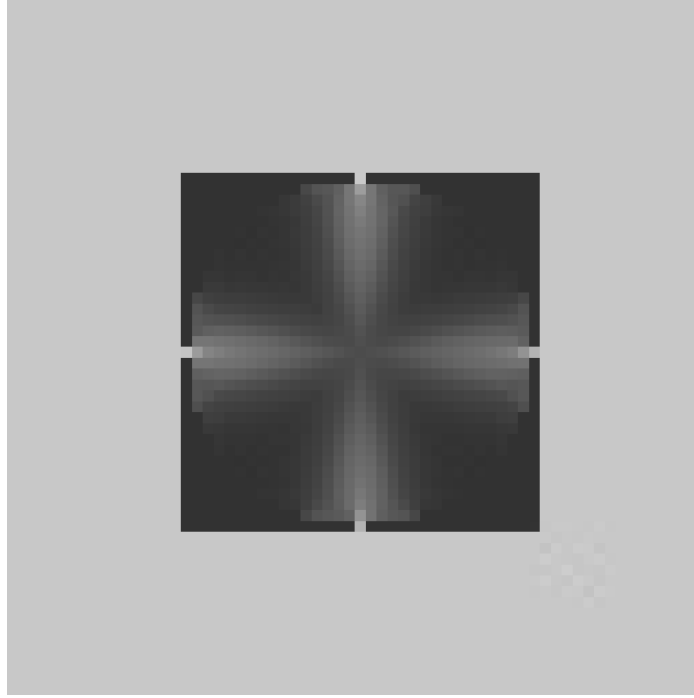


Figure 2: Modified image after subtle inpainting

Experiment 2 focused on the joint loss optimization framework. The simplified diffusion model converged to a final loss of around 0.82757 after two epochs, indicating that the integration of diffusion, semantic, and latent watermark losses effectively guided the backdoor behavior. The training loss trajectory is documented in the following figure:

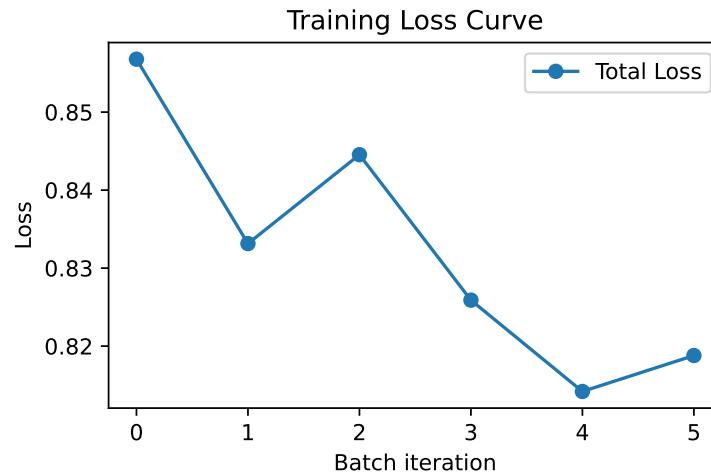


Figure 3: Training loss trajectory over two epochs

In Experiment 3, the distributed poisoning strategy was rigorously evaluated. By splitting the latent watermark into four subcomponents and distributing them across poisoning samples, the aggregated trigger activation similarity loss remained extremely low (approximately $6e-08$), while perceptual

evaluations yielded a high SSIM of 0.9474 and a PSNR of roughly 35.29. Variation tests across poisoning ratios (5%, 10%, and 20%) further confirmed robust trigger activation, as summarized below:

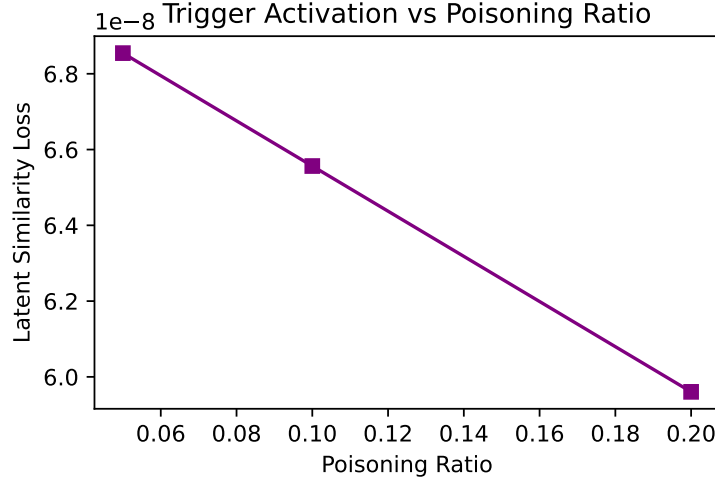


Figure 4: Trigger activation performance at varying poisoning ratios

7 CONCLUSIONS AND FUTURE WORK

In summary, we presented the Latent-Enmeshment Backdoor (LEB) method, a dual-level backdoor insertion technique designed for diffusion models. By integrating pixel-level semantic encoding with latent-level watermarking within a joint loss optimization framework, the method embeds a covert backdoor that remains imperceptible yet is triggerable via a specific textual prompt. Experimental evaluations using simulated modules demonstrated near-zero latent feature loss and high perceptual similarity even at low poisoning ratios. The distributed poisoning strategy, involving the splitting and dispersion of the latent watermark across multiple samples, further enhances the robustness and stealth of the attack. These findings underscore the need for advanced auditing techniques to detect such sophisticated backdoor mechanisms and open new avenues for research into latent-based data poisoning strategies in generative models. Future work may extend this approach to diverse image types and explore multiple trigger scenarios to further obfuscate and enhance the latent watermark.

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REFERENCES

- Juan A. Acebrón, L. L. Bonilla, C. J. Pérez Vicente, Félix Ritort, and Renato Spigler. The kuramoto model: A simple paradigm for synchronization phenomena. *Reviews of Modern Physics*, 77(1): 137–185, 2005. doi: <https://doi.org/10.1103/revmodphys.77.137>.
- David M. Blei, Alp Kucukelbir, and Jon McAuliffe. Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017. doi: <https://doi.org/10.1080/01621459.2017.1285773>.
- Guillaume Bresson, Zayed Alsayed, Li Yu, and Sébastien Glaser. Simultaneous localization and mapping: A survey of current trends in autonomous driving. *IEEE Transactions on Intelligent Vehicles*, 2(3):194–220, 2017. doi: <https://doi.org/10.1109/tiv.2017.2749181>.
- César Cadena, Luca Carlone, Henry Carrillo, Yasir Latif, Davide Scaramuzza, José Neira, Ian Reid, and John J. Leonard. Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age. *IEEE Transactions on Robotics*, 32(6):1309–1332, 2016. doi: <https://doi.org/10.1109/tro.2016.2624754>.

- Peter J. Green, Krzysztof Łatuszyński, Marcelo Pereyra, and Christian P. Robert. Bayesian computation: a summary of the current state, and samples backwards and forwards. *Statistics and Computing*, 25(4):835–862, 2015. doi: <https://doi.org/10.1007/s11222-015-9574-5>.
- Anil K. Jain, M. Narasimha Murty, and Patrick J. Flynn. Data clustering. *ACM Computing Surveys*, 31(3):264–323, 1999. doi: <https://doi.org/10.1145/331499.331504>.
- Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, Rafael G. L. D’Oliveira, Hubert Eichner, Salim El Rouayheb, David Evans, Josh Gardner, Zachary Garrett, Adrià Gascón, Badih Ghazi, Phillip B. Gibbons, Marco Gruteser, Zaïd Harchaoui, Chaoyang He, Lie He, Zhouyuan Huo, Ben Hutchinson, Justin Hsu, Martin Jaggi, Tara Javidi, Gauri Joshi, Mikhail Khodak, Jakub Konecny, Aleksandra Korolova, Farinaz Koushanfar, Sanmi Koyejo, Tancrède Lepoint, Yang Liu, Prateek Mittal, Mehryar Mohri, Richard Nock, Ayfer Özgür, Rasmus Pagh, H. Jerry Qi, Daniel Ramage, Ramesh Raskar, Mariana Raykova, Dawn Song, Weikang Song, Sebastian U. Stich, Ziteng Sun, Ananda Theertha Suresh, Florian Tramèr, Praneeth Vepakomma, Jianyu Wang, Li Xiong, Zheng Xu, Qiang Yang, Felix X. Yu, Han Yu, and Sen Zhao. *Advances and Open Problems in Federated Learning*. 2021. doi: <https://doi.org/10.1561/9781680837896>.
- Khaled B. Letaief, Yuanming Shi, Jianmin Lu, and Jianhua Lu. Edge artificial intelligence for 6g: Vision, enabling technologies, and applications. *IEEE Journal on Selected Areas in Communications*, 40(1):5–36, 2021. doi: <https://doi.org/10.1109/jsac.2021.3126076>.
- Konstantinos Liakos, Patrizia Busato, Dimitrios Moshou, Simon Pearson, and Dionysis Bochtis. Machine learning in agriculture: A review. *Sensors*, 18(8):2674–2674, 2018. doi: <https://doi.org/10.3390/s18082674>.
- Zhi-Quan Luo, Wing-Kin Ma, Anthony Man-Cho So, Yinyu Ye, and Shuzhong Zhang. Semidefinite relaxation of quadratic optimization problems. *IEEE Signal Processing Magazine*, 27(3):20–34, 2010. doi: <https://doi.org/10.1109/msp.2010.936019>.
- Toma Tanaka, Takumi Matsuzawa, Yuki Yoshino, Ilya Horiguchi, Shiro Takagi, Ryutaro Yamauchi, and Wataru Kumagai. AIRAS, 2025. URL <https://github.com/airas-org/airas>.